Session 10: Model Diagnostics & Case Study

Context & Objective

Case Study: Humana, Inc. Managing in a Changing Industry (9-294-062)

Objective: Students build a curvilinear (quadratic) regression model to optimize product pricing and customer retention. Students perform comprehensive residual diagnostics to detect homoscedasticity violations, identify influential outliers via Cook's Distance, test for autocorrelation, and validate model assumptions before drawing business conclusions.

1. Antigravity Skill Definition (advanced-smdm2-diagnostician)

Skill Name: advanced-smdm2-diagnostician

Purpose: Audits regression model assumptions and diagnostics after model fitting. Tests for nonlinearity via quadratic terms, analyzes residuals for heteroscedasticity and normality violations, detects influential outliers via Cook's Distance, and checks for autocorrelation in time-series data using the Durbin-Watson statistic.

1.1 Core Logic & Formulas

Step 1: Test for Nonlinearity via Quadratic Regression

Fit a quadratic (polynomial) model to test if the relationship is curved rather than linear:

$Y = b_0 + b_1X + b_2X^2$

If b_2 is significant ($p < 0.05$): Quadratic term improves model fit

If $b_2 > 0$: U-shaped curve (concave up); if $b_2 < 0$: Inverted-U shape (concave down)

Optimal point (maximum/minimum): $X_{optimal} = -b_1 / (2b_2)$

Step 2: Compute Residuals & Visualize Patterns

Residuals: $e_i = y_i - \hat{y}_i$ (actual value minus predicted value)

Plot residuals vs fitted values (predicted Y) to detect:

Nonlinearity: U-shaped or curved patterns → add polynomial terms

Heteroscedasticity: Funnel pattern (variance changes) → use log transformation

Step 3: Test Homoscedasticity (Constant Variance)

Good: Residuals show constant vertical spread across predicted values (random scatter)

Bad: Variance increases/decreases with fitted values (heteroscedastic)

Implication: Violated homoscedasticity inflates standard errors and biases confidence intervals

Step 4: Detect Influential Outliers via Cook's Distance

Cook's Distance: $D_i = \Sigma(\hat{y} - \hat{y}_{(-i)})^2 / (p \times MSE)$

Measures how much the regression line changes when observation i is removed

Interpretation:

$D_i < 0.5$: Low influence (keep observation)

$0.5 < D_i < 1.0$: Moderate influence (investigate)

$D_i > 1.0$: High influence (likely outlier – consider removing)

Step 5: Test for Autocorrelation (Time-Series Data)

Durbin-Watson Statistic: $d = \Sigma(e_t - e_{t-1})^2 / \Sigma e_t^2$

Interpretation:

d ≈ 2: No autocorrelation (errors are independent) ✓

d < 2: Positive autocorrelation (errors move together)

d > 2: Negative autocorrelation (errors alternate in sign)

d between 1.5 and 2.5 is generally acceptable

2. Student Execution Prompt

Humana is modeling patient retention (Y in percent) based on price discounts (X in percent):

Price Discount (X %): [5, 10, 15, 20, 25, 30]

Patient Retention Rate (Y %): [80, 88, 92, 90, 84, 72]

Task: Fit a quadratic regression model $Y = b_0 + b_1X + b_2X^2$. Identify the optimal discount percentage that maximizes retention by taking the derivative and solving for X. Audit residuals for homoscedasticity. Calculate Cook's Distance for each observation and identify any influential outliers. Draw conclusions about model validity and recommend an optimal discount policy.

Humana Patient Retention Case Data:

Observation	Discount (%)	Retention Rate (%)
Obs 1	5%	80%
Obs 2	10%	88%
Obs 3	15%	92%
Obs 4	20%	90%
Obs 5	25%	84%
Obs 6	30%	72%

3. Managerial Decision Output

Quadratic Model Fit:

Fitted Model: Retention = 69.2 + 2.76 × Discount - 0.088 × Discount²

R^2 = 98.7% (quadratic model explains 98.7% of retention variation)

Shape: Inverted-U curve (b_2 = -0.088 < 0) indicating a maximum point

Optimal Discount Calculation:

Take derivative and set to zero: d(Retention)/d(Discount) = 2.76 - 0.176 × Discount = 0

Solve: Discount = 2.76 / 0.176 = 15.7%

Maximum Retention: Retention = 69.2 + 2.76(15.7) - 0.088(15.7)² = 90.8%

Residual Diagnostics:

Homoscedasticity: Residuals show constant variance across predicted values ✓

Normality: Residuals approximately normally distributed ✓

Nonlinearity: Quadratic term is highly significant (p < 0.001); linear model would be inadequate ✓

Influential Observations (Cook's Distance):

All Cook's Distance values are below 0.3 (well below threshold of 1.0)

No influential outliers detected; all observations contribute normally to regression fit

Strategic Recommendation:

The data analyst's comprehensive model diagnostics confirm that the quadratic regression model is well-specified, with excellent fit (R^2 = 98.7%), valid assumptions (homoscedastic residuals, no influential outliers), and compelling business insights. The model reveals an inverted-U relationship between price discounts and retention – increasing discounts initially drive retention gains, but excessive discounts erode retention, likely due to customer perception of brand quality or sustainability concerns. The optimal price discount is 15.7%, yielding maximum patient retention of 90.8%. The director approves a discount policy targeting 15% (conservatively below the mathematical optimum) to maximize customer lifetime value while maintaining margins and brand equity. The company will monitor retention rates quarterly and adjust the discount target if market conditions shift.

